

شرح المادة
التقنيات العددية

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chapter: .....

مادة التقنيات العددية

Subject: Numerical Techniques



1] root of equation

2] Matrix

3] Curve fitting (regression)

4] Interpolation

5] Derivatives and integration

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chapter: .....

مادة التقنيات العددية

Subject: Curve fitting

Least square Regression

1] Linear Regression

2] 2nd order Polynomial Regression

3] Multi Linear Regression

4] Exponential Regression

* فكرة هذا الجزء من المادة انو يعطينا عدة مشاهدات لمتغيرين (متغير بيندا) بحيث نجد منها اقرب تعبير عن العلاقة بينهما و يتل هذه المشاهدات عن طريق ايجاد مقياس الخطأ كالتالي:-

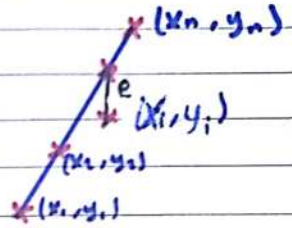
$$S = \sum_{i=1}^{n} (y_i - F(x_i))^2$$

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linear Regression $(F(x) = ax + b)$

Given n {observation Points} $(x_1, y_1)(x_2, y_2) \dots (x_n, y_n)$
then we want to find best linear function $F(x)$

$$S = \sum_{i=1}^{n} (y_i - F(x_i))^2$$
$$= \sum_{i=1}^{n} (y_i - (ax_i + b))^2$$



الهدف: ايجاد اقرب خط اقرب ما يمكن للنقاط ، بحيث انه يعطى هو (x, y) والمطلوب هو (a, b)

∴ solve $\frac{\partial S}{\partial a} = 0$, $\frac{\partial S}{\partial b} = 0$ } critical Point

$$\therefore \frac{\partial S}{\partial a} = \sum_{i=1}^{n} 2 (y_i - (ax + b))(-x) = 0$$

$$\frac{\partial S}{\partial b} = \sum_{i=1}^{n} 2 (y_i - (ax + b))(-1) = 0$$

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تذكر

$$\sum_{i=1}^{n}(a+b) = \sum_{i=1}^{n}a + \sum_{i=1}^{n}b \quad ; \quad \sum_{i=1}^{n}cx_i = c\sum_{i=1}^{n}x_i$$

$$\sum_{i=1}^{n}1 = n \quad ; \quad \sum_{i=1}^{n}c = cn \text{ , where } c \text{ is constant}$$

$$\bar{x} = \frac{\sum_{i=1}^{n}x_i}{n}$$

$$\therefore \sum_{i=1}^{n}y_i x_i - a\sum_{i=1}^{n}x_i^2 - b\sum_{i=1}^{n}x_i = 0 \quad \dots (1)$$

$$\hookrightarrow b\Sigma x_i + a\Sigma x_i^2 = \Sigma y_i x_i$$

$$\sum_{i=1}^{n}y_i - a\sum_{i=1}^{n}x_i - b\sum_{i=1}^{n}1 = 0 \quad \dots (2)$$

$$\hookrightarrow bn + a\Sigma x_i = \Sigma y_i$$

* المعادلات مع المجهول تسمى Normal equations

$$\begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix} \begin{bmatrix} b \\ a \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum y_i x_i \end{bmatrix}$$

A $\quad$ X $\quad$ b

$\therefore$ symetric Matrix (دائماً) $\rightsquigarrow A = A^T$

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بعد ترتيب المعادلات نحصل على:

حيث:

$$* \; a = \frac{n \Sigma x_i y_i - \Sigma x_i \Sigma y_i}{n \Sigma x_i^2 - (\Sigma x_i)^2}$$

$$* \; b = \bar{y} - a \bar{x}$$

$$\bar{y} = \frac{\Sigma y_i}{n}$$

$$\bar{x} = \frac{\Sigma x_i}{n}$$

او نستطيع ايجاد هم عن طريق الله الحاسبة بادخال المعادلات 1 و 2

* نحصل بعدها على القرانة $F(x) = ax + b$ والذي يمثل هذه المشاهدات فلو طلب توقع نتيجة بالنسبة للاخرى نجدها من خلال القرانة مثلاً لو طلب ايجاد $f(2)$ .. حيث تكون $(x = 2)$ و y نجدها من خلال القرانة اي اصبح بامكاننا توقع مشاهدات اخرى

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example :- Fit a straight Line to the x and y value in the following table

x_i	1.1	2	3.2	4	4.5	5.2
y_i	0.5	2.5	2	4	3.5	6.5

Solution: $F(x) = ax + b$ ✱

$S = \sum_{i=1}^{n} (y_i - ax - b)^2$

$$a = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}$$

$= 1.1734$

$$b = \bar{y} - a\bar{x}$$

$\bar{y} = \frac{19}{6} = 3.1667$

$\bar{x} = \frac{20}{6} = 3.333$

$= -0.7446$

x_i	y_i	x_i^2	$x_i y_i$
1.1	0.5	1.21	0.55
2	2.5	4	5
3.2	2	10.24	6.4
4	4	16	
4.5	3.5		
5.2	6.5		
Σ 20	19	78.74	77.5

$F(x) = 1.1734x - 0.7446$

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Polynomial Regression

$F(x) = ax^2 + bx + c$ ... 2nd order Poly.

$$S = \sum_{i=1}^{n} (y_i - ax_i^2 - bx_i - c)^2$$

$$\frac{\partial S}{\partial a} = \sum_{i=1}^{n} 2(y_i - ax_i^2 - bx_i - c)(-x_i^2) = 0 \quad \dots [1]$$

$$\Rightarrow c\sum x_i^2 + b\sum x_i^3 + a\sum x_i^4 = \sum y_i x_i^2$$

$$\frac{\partial S}{\partial b} = \sum_{i=1}^{n} 2(y_i - ax_i^2 - bx_i - c)(-x_i) = 0 \quad \dots [2]$$

$$\Rightarrow c\sum x_i + b\sum x_i^2 + a\sum x_i^3 = \sum y_i x_i$$

$$\frac{\partial S}{\partial c} = \sum_{i=1}^{n} 2(y_i - ax_i^2 - bx_i - c)(-1) = 0 \quad \dots [3]$$

$$\Rightarrow c(n) + b\sum x_i + a\sum x_i^2 = \sum y_i$$

* المعادلات مع بعضها تسمى Normal equation

* عند الاشتقاق الجزئي بالنسبة لـ (a) نعتبر (a) متغير و الباقي ثوابت وكذلك الامر بالنسبة لي (c , b)

Symmetric $A = A^T$

$$\begin{bmatrix} n & \sum x_i & \sum x_i^2 \\ \sum x_i & \sum x_i^2 & \sum x_i^3 \\ \sum x_i^2 & \sum x_i^3 & \sum x_i^4 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum y_i x_i \\ \sum y_i x_i^2 \end{bmatrix}$$

$$A \cdot X = B$$

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example :- Fit a second order polynomial to the x and y values in the following, $y = ax^2 + bx + c$

x_i	1.1	2	3.2	4	4.5	5.2
y_i	0.5	2.5	2	4	3.5	6.5

, then Find F(6)

solû:

$$\begin{bmatrix} 6 & 20 & 78.74 \\ 20 & 78.74 & 337.3 \\ 78.74 & 337.7 & 1519.53 \end{bmatrix} \begin{bmatrix} c \\ b \\ a \end{bmatrix} = \begin{bmatrix} 19 \\ 77.5 \\ 341.72 \end{bmatrix}$$

x_i	y_i	x_i^2	x_i^3	x_i^4	$x_i y_i$	$x_i^2 y_i$
1.1	0.5	1.21	1.33	1.46	0.55	0.61
2	2.5	4	8	16	5	10
3.2	2	10.24	32.68	104.85	64	20.48
4	4	16	64	256	16	64
4.5	3.5	20.25	91.12	410.06	15.75	70.8
5.2	6.5	27.04	140.6	731	33.8	175.7
$\Sigma =$ 20	19	78.74	337.7	1519.3	77.5	341.7

$\therefore$ 6c + 20b + 78.74a = 19

20c + 78.74b + 337.3a = 77.5

78.74c + 337.3b + 1519.5a = 341.7

$\therefore$ apply Cramer's rule

$\therefore$ $a = 0.239 \quad b = -0.315 \quad c = 1.084$

$F(x) = y = 0.239x^2 - 0.315x + 1.084$

$F(6) = 0.239(6)^2 - 0.315(6) + 1.084 \ldots \square$

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example:- For 5 Points $[(x_i, y_i)]_{i=1}^{5}$ fit these Points with least the following data :-

$\sum_{i=1}^{5} x_i = 2.5$, $\sum_{i=1}^{5} x_i^2 = 1.875$, $\sum_{i=1}^{5} x_i^3 = 1.5625$

$\sum y_i = 8.768$, $\sum x_i y_i = 5.4514$, $\sum x_i^2 y_i = 4.4018$

$\sum x_i^4 = 1.3828$

* linear system.

sol$\hat{u}$:-

$$5a_1 + 2.5a_2 + 1.875a_3 = 8.768$$
$$2.5a_1 + 1.875a_2 + 1.5625a_3 = 5.4514$$
$$1.8759a_1 + 1.5625a_2 + 1.3828a_3 = 4.4081$$

$a_1 = 1.0051$ $\Big\{$ $a_2 = 0.86468$ $\Big\{$ $a_3 = 0.84316$

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Multi Linear Regression

- The equation $ax+b=0$ is linear with one variable
- The equation $ax_1+bx_2+d=0$ is linear with two variable
- The equation $a_1x_1+a_2x_2+\dots+a_nx_n+b=0$ is linear with n-variable

↝ To fit the data $(x_i, y_i, z_i), \dots (x_n, y_n, z_n)$ by a linear Function with two variable $z = ax+by+d$

$$S = \sum_{i=1}^{n} (z_i - (ax+by+d))^2$$

$$\frac{\partial S}{\partial a} = \sum_{i=1}^{n} 2(z_i - ax_i - by_i - d)(-x_i) = 0 \quad \rightsquigarrow d\Sigma x_i + b\Sigma x_i y_i + a\Sigma x_i^2 = \Sigma z_i x_i \dots (1)$$

$$\frac{\partial S}{\partial b} = \sum_{i=1}^{n} 2(z_i - ax_i - by_i - d)(-y_i) = 0 \quad \rightsquigarrow d\Sigma y_i + b\Sigma y_i^2 + a\Sigma x_i y_i = \Sigma z_i y_i \dots (2)$$

$$\frac{\partial S}{\partial d} = \sum_{i=1}^{n} 2(z_i - ax_i - by_i - d)(-1) = 0 \quad \rightsquigarrow d(n) + b\Sigma y_i + a\Sigma x_i = \Sigma z_i \dots (3)$$

$$\begin{bmatrix} n & \Sigma y_i & \Sigma x_i \\ \Sigma y_i & \Sigma y_i^2 & \Sigma x_i y_i \\ \Sigma x_i & \Sigma y_i x_i & \Sigma x_i^2 \end{bmatrix} \begin{bmatrix} d \\ b \\ a \end{bmatrix} = \begin{bmatrix} \Sigma z_i \\ \Sigma z_i y_i \\ \Sigma z_i x_i \end{bmatrix}$$

$$A \cdot X = B$$

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Ex: Fit a linear Function with two variable by $(z = ax + by + c)$, the data is

معطى — انت بدك توجدهم

x_i	y_i	z_i	$x_i y_i$	y_i^2	x_i^2	$z_i y_i$	$z_i x_i$
1	2	1	2	4	1	2	1
3	4	2	12	16	9	8	6
5	6	3	30	36	25	18	15
7	8	4	56	64	49	32	28
Σ 16	20	10	100	120	84	60	50

$$\begin{bmatrix} n & \Sigma y_i & \Sigma x_i \\ \Sigma y_i & \Sigma y_i^2 & \Sigma y_i x_i \\ \Sigma x_i & \Sigma x_i y_i & \Sigma x_i^2 \end{bmatrix} \begin{bmatrix} c \\ b \\ a \end{bmatrix} = \begin{bmatrix} \Sigma z_i \\ \Sigma z_i y_i \\ \Sigma z_i x_i \end{bmatrix}$$

$$\begin{bmatrix} 4 & 20 & 16 \\ 20 & 120 & 100 \\ 16 & 100 & 84 \end{bmatrix} \begin{bmatrix} c \\ b \\ a \end{bmatrix} = \begin{bmatrix} 10 \\ 60 \\ 50 \end{bmatrix}$$

$$\begin{aligned} 4c + 20b + 16a &= 10 \\ 20c + 120b + 100a &= 60 \\ 16c + 100b + 84a &= 50 \end{aligned}$$

لو بدك تحلها على الآلة الحاسبة ما رح يطلع جواب لانه الكتب اسئلتها القيم غير معقدة بالرقم أنت افهم الفكرة ورح احلها ارتحام عشان بعدين بسهلك الكتب الحاسبة

$$z = 4x + 3y + 1$$

مثلا بدو $F(3,2)$

$$F(3,2) = 4*3 + 3*2 + 1 = 19$$

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Exponential Regression

→ Give $(x_1, y_1) \dots (x_n, y_n)$

→ $F(x) = a e^{bx} = y$

$$S = \sum_{i=1}^{n} (y_i - F(x))^2$$

∴ $\ln y = \ln (a e^{bx})$

$= \ln a + \ln e^{(bx)}$

$= \underbrace{\ln a}_{a_1 \text{ نفرضها}} + bx$

$\ln y = bx + a_1$

then solve:

$n a_1 + b \Sigma x = \Sigma \ln y$

$a_1 \Sigma x + b \Sigma x^2 = \Sigma \ln y \ *$

] نجد a_1 & b عن طريق المعادلتين الخاصة

$a_1 = \ln a \dots \curvearrowright a = e^{a_1}$

∴ $y = a e^{bx}$

هكذا نكون قد اوجدنا a & b

ثم بحمد الله

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chapter: .....

مادة التقنيات العددية

Subject: Interpolation

1] Interpolation By Newton divided Difference Polynomial

2] lagrange Interpolation

الهدف هو ايجاد كثير حدود يعبر عن مشاهدات معطاه بحيث نحصل منه على اقترانه كثير حدود يمثل افضل تقريب للاقترانه

* نقاط (مشاهدات) ← اقترانه كثير حدود
اقترانه (ومعطى عليه نقاط)

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#Interpolation By Newton divided difference Polynomial

let $(x_0, y_0) \dots (x_n, y_n)$ nods where $y_0 = F(x_0)$ To Find Poly. $P_n(x)$ such that:-

$F(x) = P_n(x)$ & $F(x_i) = P_n(x_i) = y_i$

$P_n(x) = b_0 + b_1(x-x_0) + b_2(x-x_0)(x-x_1) + b_3(x-x_0)(x-x_1)(x-x_2) \dots$
$+ b_n(x-x_0)(x-x_1)\dots(x-x_{n-1})$

نسمى هذه هي معادلة N.D.D

* Not that :-

$F(x_0) = P_n(x_0) = b_0 + 0 \dots \rightarrow \underline{b_0 = F(x_0)}$

$F(x_1) = P_n(x_1) = b_0 + b_1(x_1 - x_0) \dots \rightarrow b_1 = \underline{\frac{F(x_1) - F(x_0)}{x_1 - x_0}}$

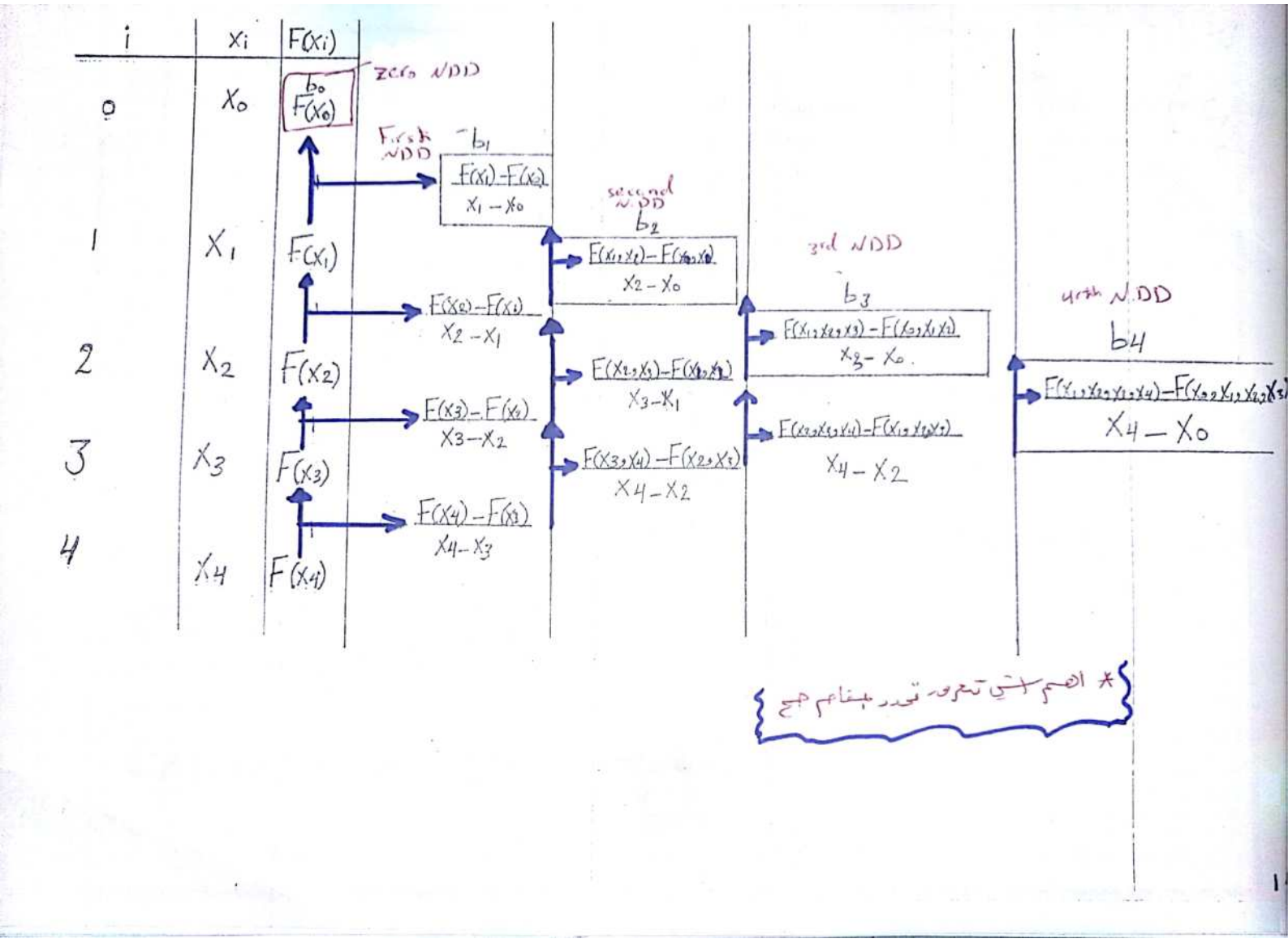
$F(x_2) = F(x_0) + \frac{F(x_1) - F(x_0)}{x_1 - x_0}(x_1 - x_0) + b_2(x_2 - x_0)(x_2 - x_1)$

$\therefore F(x_2) = F(x_0) + F(x_1) - F(x_0) + b_2(x_2 - x_0)(x_2 - x_1)$

$\therefore b_2 = \underline{\frac{F(x_2) - F(x_1)}{(x_2 - x_0)(x_2 - x_1)}}$

* هذا هو استنتاج القوانين
ويمكنه اعتماد الحل يكونه باستخدام
Interpolation في الصفحة التالية

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example :- Given $(x_0, x_1, x_2) = (1, 1.1, 1.2)$ lie on $F(x) = \tan(x)$, Find 2nd order Divided Difference interpolation poly. , to approximate $F(1.15)$

soln:-

i	x_i	$f(x_i)$ & b_0	& b_1	& b_2
0	1	1.5574	4.074	$\frac{6.674 - 4.074}{1.2 - 1} = 10$
1	1.1	1.9648	6.674	
2	1.2	2.5722		

$$P_2(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1)$$

$$= 1.5574 + 4.074(x - 1) + 10(x - 1)(x - 1.1)$$

$$F(1.15) \approx P_2(1.15)$$
$$\approx 2.2435$$

$$F(1.15) = \tan(1.15) = 2.2344$$

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Example For the nods $(1,3)^{x_0, y = F(x_0)}$, $(2,5)$, $(-1,6)$, $(8,4)$
Find Newton divided difference Poly.

solû:-

i	x_i	$F(x_i)$	b_1	b_2	b_3
0	1	3 (b_0)	2	7/6	
1	2	5			-31/189
2	-1	6	-1/3		
3	8	4	-2/9	1/54	

$$P_3(x) = 3 + 2(x-1) + 7/6(x-1)(x-2) - 31/189\,(x-1)(x-2)(x+1)$$

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Lagrange Interpolation

- Given set of Point $(x_0, F(x_0))\ (x_1, F(x_1)) \dots (x_n, F(x_n))$

n+1 point

- Approximately F By a polynomial (Lagrange poly.) $P_n(x)$ of degree n ⟶ $F(x) \approx P_n(x)$

- General form of Lagrange poly. is:-

$$P_n(x) = \sum_{i=1}^{n} F(x_i)\, L_i(x) = f(x_0)L_0(x) + f(x_i)L_i(x) \dots + f(x_n)L_n(x)$$

where:-

$$L_i(x) = \prod_{\substack{k=0 \\ k \neq i}}^{n} \frac{x - x_k}{x_i - x_k}$$

(تجرعة بالضرب)

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Example:- Find Lagrange poly. for the following:-

1] Givin $(X_0, F(x_0))$, $(X_1, F(x_1))$ ... 2 Point

sol:-

x	X_0	X_1
y	$F(X_0)$	$F(X_1)$

$$P(X) = F(X_0)\,L_0(X) + F(X_1)\,L_1(X)$$

$$= F(X_0)\cdot\frac{X-X_1}{X_0-X_1} + F(X_1)\,\frac{X-X_0}{X_1-X_0}$$

Lagrange poly. of degree one ≈ the Same NDD poly of degree one (equal $F(X_0)$)

2] Givin $(X_0, F(X_0))$, $(X_1, F(X_1))$, $(X_2, F(X_2))$ ... 3 Points

x	X_0	X_1	X_2
y	$F(X_0)$	$F(X_1)$	$F(X_2)$

$$P_2(X) = F(X_0)\,L_0(X) + F(X_1)\,L_1(X) + F(X_2)\,L_2(X)$$

$$= F(X_0)\frac{(X-X_1)(X-X_2)}{(X_0-X_1)(X_0-X_2)} + F(X_1)*\frac{(X-X_0)(X-X_2)}{(X_1-X_0)(X_1-X_2)}$$

$$+ F(X_2)\frac{(X-X_0)(X-X_1)}{(X_2-X_0)(X_2-X_1)}$$

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example:- Use the numbers $x_0 = 2$ $x_1 = 2.75$ $x_2 = 4$ to Find 2nd order lagrange poly. for $F(x) = \frac{1}{x}$ to approximate $F(3)$

Solû:-

x	2	2.75	4
$F(x)$	1/2	4/11	1/4

$$P_n(x) = 1/2 * \frac{(x-2.75)(x-4)}{(2-2.75)(2-4)} + 4/11 * \frac{(x-2)(x-4)}{(2.75-2)(2.75-4)}$$

$$+ 1/4 * \frac{(x-2)(x-2.75)}{(4-2)(4-2.75)}$$

$$= \frac{1}{22}x^2 - \frac{35}{88}x + \frac{49}{44}$$

$$F(3) \approx P(3) = 0.32955$$

بسم الله

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chapter: .....

مادة التقنيات العددية

Subject: Derivatives & Integration

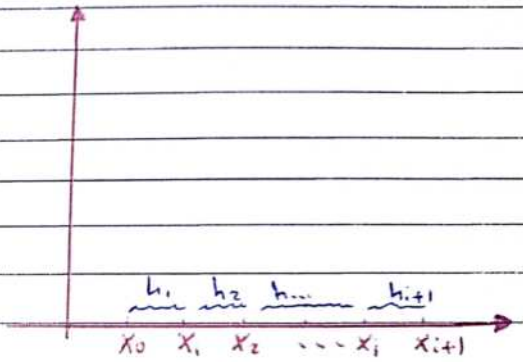
Equally Spaced Partition

Recall: [a,b] Partition For [a,b]

$a = x_0 , x_1 , x_2 \dots x_n = b$

where: $x_i < x_{i+1}$ $\forall$ $i = 0 \dots n-1$

$h_i = x_{i+1} - x_i$ $\forall$ $i = 0, 1 \dots n-1$



if: $h_i = h_j$ $\forall$ $i, j = 1, 2 \dots n$

then the Partition is called equally spaced Partition

* اي انه الفروقات بينه النقاط متساوين

$x_i = x_0 + ih$

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Numerical Differentiation تقريب بالاشتقاقات

Remember: The taylorpoly. For $F(x)$ at Point x_{i+1} is

$$F(x) \approx P_n(x_i) = F(x_i) + F'(x_i)(x - x_i) + \frac{F''(x_i)(x - x_i)^2}{2!} + \dots$$

at Point x_{i+1} :-

$$F(x_{i+1}) = F(x_i) + F'(x_i)(x_i + h - x_i) \quad \dots [1]$$

$$\hookrightarrow F'(x_i) = \frac{F(x_{i+1}) - F(x_i)}{h} \quad \dots > \text{Foreword divided difference}$$

at Point x_{i-1} :-

$$F(x_{i-1}) = F(x_i) + F'(x_i)(x_i - h - x_i) \quad \dots [2]$$

$$\hookrightarrow F'(x_i) = \frac{F(x_i) - F(x_{i-1})}{h} \quad \dots > \text{Backword divided difference}$$

where allways $h > 0$

اذا كانت النقاط بعطاه موحيه

$$h = x_{i+1} - x_i$$

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* By Equation 1 & 2 we got:

equa 1 : $F(x_i) = F(x_i+h) - h F'(x_i)$

equa 2: $F(x_i) = hF'(x_i) + F(x_i-h)$

عندما نساوي المعادلتين مع بعض:

$$F(x_i+h) - h F'(x_i) = h F'(x_i) + F(x_i-h)$$

$$F(x_i+h) - F(x_i-h) = 2h F'(x_i)$$

$$\Rightarrow F'(x_i) = \frac{F(x_i+h) - F(x_i-h)}{2h}$$ ...> Centered divided difference

نذكر

always $h > 0$

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Example:- $F(x) = -0.1x^4 - 0.15x^3 - 0.5x^2 - 0.25x + 1.2$

approximate $F'(0.5)$ using Finite divided difference method with step size $h = 0.25$

Solution:-

x	0.25	0.5	0.75
F(x)	□	□	□

Forward ...> $f'(0.5) = \frac{F(0.75) - F(0.5)}{0.25} = -1.155$

Backward ...> $f'(0.5) = \frac{F(0.5) - F(0.25)}{0.25} = -0.714$

Centered ...> $f'(0.5) = \frac{F(0.75) - F(0.25)}{2(0.25)} = -0.934$ ادق

Real Value $f'(0.5) = -0.9125$

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Example:- For the nods $(\overset{x_0}{0}, 2)$, $(\overset{x_1}{1}, 5)$, $(\overset{x_2}{2}, 9)$ approximate $F'(1)$ By Forword & Backword divided difference.

solu: $h = 1 - 0 = 1$

1— Forword solu.:

$$f'(1) = \frac{F(2) - F(1)}{1} = 9 - 5 = 4$$

2- Backword solu:

$$f'(1) = \frac{-f(0) + f(1)}{1} = -2 + 5 = 3$$

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Moreover:

لإيجاد المشتقة الثانية

$$F(x) \approx F(x_i) + F'(x_i)(x-x_i) + F''(x_i)\frac{(x-x_i)^2}{2!} + F'''(x_i)\frac{(x-x_i)^3}{3!}$$

at Point (x_{i+1}):

$$F(x_{i+1}) = F(x_i) + F'(x_i)\overbrace{(x_{i+h}-x_i)}^{h} + F''(x_i)\frac{h^2}{2} + F'''(x_i)\frac{h^3}{3!} \quad \cdots [1]$$

at Point (x_{i-1}):

$$F(x_{i-1}) = F(x_i) + F'(x_i)*-h + F''(x_i)\frac{h^2}{2} + F'''(x_i)*\frac{-h^3}{3!} \quad \cdots [2]$$

* نجعل في كل معادلة $F(x_i)$ موضوع القانون ثم نساوي المعادلتين ببعض فنحصل على: ← ← ←

$$F(x_{i+h}) + F(x_{i-h}) = 2F(x_i) + F''(x_i)h^2$$

$$\vdots$$

$$\rightsquigarrow F''(x_i) = \frac{F(x_{i+h}) + F(x_{i-h}) - 2F(x_i)}{h^2}$$

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Derivative By Lagrange interpolation

For the partition $x_0, x_1, x_2 \dots x_n$, Equally spaced with step size h

$$F(x) \approx P_n(x) = \sum_{i=1}^{n} F(x_i) L_i(x) \quad \dots \text{Lagrange interpolation}$$

$$F'(x) = \left[\sum_{i=1} F(x_i) L_i(x)\right]' = \sum_{i=1} \left[F(x_i) L_i(x)\right]' = \sum_{i=1} F(x_i) L_i'(x) \dots$$

(n+1) Point Formula to appro. $F'(x_i)$

Example: Consider the nods $(1, 2)$ (x_0), $(3, 5)$ (x_1), $(4, 7)$ (x_2) Find $F'(1)$. By using second Lagrange interpolation

هذا المثال فقط لتوضيح آلية اشتقاق القوانين لكن في الامتحان سنستخدم القوانين المشتقة لاحقاً

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solû: $F'(x) = \sum_{i=1}^{n} F(x_i) L_i'(x)$ / $(\overset{x_0}{1}, 2)$, $(\overset{x_1}{3}, 5)$, $(\overset{x_2}{4}, 7)$ / $F'(1) = ?$

$$L_0(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} = \frac{(x-3)(x-4)}{(1-3)(1-4)} = \frac{1}{6}(x^2-7x+4)$$

$$L_1(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} = \frac{(x-1)(x-4)}{(3-1)(3-4)} = \frac{-1}{2}(x^2-5x+4)$$

$$L_2(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} = \frac{(x-3)(x-1)}{(4-3)(4-1)} = \frac{1}{3}(x^2-4x+3)$$

$$P_2(x) = \overset{F(x_0)}{2} \cdot \frac{1}{6}(x^2-7x+4) + \overset{F(x_1)}{5} \cdot \frac{-1}{2}(x^2-5x+4) + \overset{F(x_2)}{7} \cdot \frac{1}{3}(x^2-4x+3)$$

$$F'(x) = \frac{1}{3}(2x-7) - \frac{5}{2}(2x-5) + \frac{7}{3}(2x-4)$$

$$F'(1) = \frac{7}{6} \quad \#$$

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Ex8 Consider the nods $(x_0, F(x_0))$, $(x_1, F(x_1))$, $(x_2, F(x_2))$ with $h = (x_1 - x_0) = (x_2 - x_1)$, Find $F'(x_0)$ by using second lagrange interpolation

$$L_0(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} = \frac{(x-x_1)(x-x_2)}{2h^2}$$

$(x_0-x_1) = -h$, $(x_0-x_2) = -2h$

$$L_1(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} = \frac{(x-x_0)(x-x_2)}{-h^2}$$

$(x_1-x_0) = h$, $(x_1-x_2) = -h$

$$L_2(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} = \frac{(x-x_0)(x-x_1)}{2h^2}$$

$(x_2-x_0) = 2h$, $(x_2-x_1) = h$

$$F(x) \approx P_2(x) = F(x_0)\frac{(x-x_1)(x-x_2)}{2h^2} + F(x_1)\frac{(x-x_0)(x-x_2)}{-h^2} + F(x_2)\frac{(x-x_0)(x-x_1)}{2h^2}$$

$$F'(x) = F(x_0)\left[\frac{2x-(x_1+x_2)}{2h^2}\right] + F(x_1)\left[\frac{2x-(x_0+x_2)}{-h^2}\right] + F(x_2)\left[\frac{2x-(x_0+x_1)}{2h^2}\right]$$

$$F'(x_0) = F(x_0)\left[\frac{2x_0-(x_1+x_2)}{2h^2}\right] + F(x_1)\left[\frac{2x_0-x_0-x_2}{-h^2}\right] + F(x_2)\left[\frac{2x_0-x_0-x_1}{2h^2}\right]$$

$x_0-x_1 = -h$, $x_0-x_2 = -2h$ $\quad$ $x_0-x_2 = -2h$ $\quad$ $x_0-x_1 = -h$

$$F'(x_0) = F(x_0)\left[\frac{-h+-2h}{2h^2}\right] + F(x_1)\left[\frac{-2h}{-h^2}\right] + F(x_2)\left[\frac{-h}{2h^2}\right]$$

$$F'(x_0) = F(x_0)*\frac{-3}{2h} + F(x_1)*\frac{2}{h} + F(x_2)*\frac{-1}{2h}$$

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$$F'(X_0) = \frac{-3}{2h} F(X_0) + \frac{2*2}{h*2} F(X_1) + \frac{-1}{2h} F(X_2)$$

$F(X_1)$ عامل مشترك $\frac{1}{2h}$ بدي أطلع

$$F'(X_0) = \frac{1}{2h}\left[-3F(X_0) + 4F(X_1) - F(X_2)\right]$$

$X_1 = X_0 + h$ و $X_2 = X_0 + 2h$

$$F'(X_0) = \frac{1}{2h}\left[-3F(X_0) + 4F(\underbrace{X_0+h}_{X_1}) - F(\underbrace{X_0+2h}_{X_2})\right] \longrightarrow 1$$

$$F'(X_1) = \frac{1}{2h}\left[F(\underbrace{X_0+2h}_{X_2}) - F(X_0)\right] \longrightarrow 2$$

$$F'(X_2) = \frac{1}{2h}\left[F(X_0) - 4F(\underbrace{X_0+h}_{X_1}) + 3F(\underbrace{X_0+2h}_{X_2})\right] \longrightarrow 3$$

* Replace X_0 in Equation 2 by $X_0 - h$

$X_0 - h$ بنحوض بدل X_0 المعادلة رقم 2

$$F'(X_0) = \frac{1}{2h}\left[F(X_0 - h + 2h) - F(X_0 - h)\right] = \frac{1}{2h}\left[F(X_0+h) - F(X_0-h)\right] \longrightarrow 2^*$$

* Replace X_0 in Equation 3 by $X_0 - 2h$

$$F'(X_2) = \frac{1}{2h}\left[F(X_0-2h) - 4F(X_0-2h+h) + 3F(X_0-2h+2h)\right]$$

$$F'(X) = \frac{1}{2h}\left[F(X_0-2h) - 4F(X_0-h) + 3F(X_0)\right] \longrightarrow 3^*$$

Equation 1 & 2* is Called three Point Formula

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Derevative By lagrange interpolation

* Three Point Formula:

بعد اشتقاقه، المعادلات نحصل على:

$$F'(x_0) = \frac{1}{2h}\left[-3F(x_0) + 4F(x_1) - F(x_2)\right] \quad \ldots\rightarrow 2$$

$$F'(x_0) = \frac{1}{2h}\left[F(x_0+h) - F(x_0-h)\right] \quad \ldots\rightarrow 2^*$$

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Example:- Consider the table, approximate $F'(0.1)$ & $F'(0.2)$ using three Point Lagrange formula, and Find $F''(0.2)$

	x_0	x_1	x_2
x_i	0.1	0.2	0.3
$F(x_i)$	2	5	9

$h = 0.2 - 0.1 = 0.1$

Solu:-

$$F'(0.1) = \frac{1}{2h}\left[-3F(x_0) + 4F(x_1) - F(x_2)\right]$$

$$= \frac{1}{2(0.1)}\left[-3(2) + 4(5) - 9\right]$$

$$= 25$$

$$f'(0.2) = \frac{1}{2h}\left[F(x_0+h) - F(x_0-h)\right]$$

$$= \frac{1}{2(0.1)}\left[9 - 2\right] = 35$$

$$f''(0.2) = \frac{F(x_{i+1}) + F(x_{i-1}) - 2F(x_i)}{h^2} = 100$$

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Numerical integration

* تذكر انه التكامل يعبر عنه بالمساحة تحت الاقتران

1] Trapezoidal Method : to approximate $\int_a^b F(x)\,dx$ where

$F(x) \approx P_n(x)$ ... $P_n(x)$ is 1st order ($n=1$) Lagrange interpolation

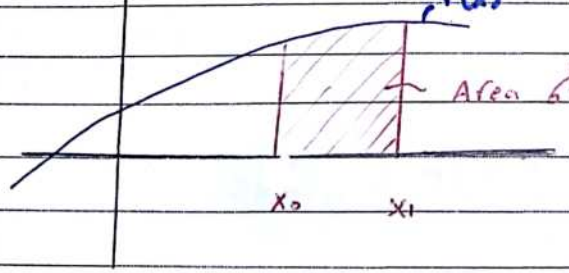
$$\therefore F(x) \approx P_1(x) = F(x_0)\frac{x-x_1}{x_0-x_1=-h} + F(x_1)\frac{x-x_0}{x_1-x_0=h}$$

$$F(x) = \frac{1}{h}\left[F(x_1)(x-x_0) - F(x_0)(x-x_1)\right], \quad h = x_1 - x_0$$

$$\therefore \int_a^b F(x)\,dx = \frac{1}{h}\left[F(x_1)\int_a^b (x-x_0)\,dx - F(x_0)\int_a^b (x-x_1)\right]$$

$$\therefore \int_a^b F(x)\,dx = \frac{h}{2}\left(F(x_1) + F(x_0)\right) \quad \text{...} \rightarrow \text{Trapezoidal Rule}$$

$h = b-a = x_1 - x_0$



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Example: use Trapezoidal Rule to approximate

$$\int_{a=\sqrt{\ln 2}}^{b=\sqrt{\ln 5}} e^{x^2}\,dx \quad \rightsquigarrow F(x) = e^{x^2}, \quad h = \sqrt{\ln 2} - \sqrt{\ln 5}$$

solu: $F(a) = e^{(\sqrt{\ln 2})^2} = 2$

$F(b) = 5$

$$\Rightarrow \int_a^b e^{x^2}\,dx = \frac{\sqrt{\ln 2} - \sqrt{\ln 5}}{2}(2+5) = 1.526$$

Example: approximate $\int_0^2 \frac{dx}{x+1}$ by Trapezoidal Rule and Find absolute (exact) error

solu:

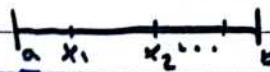
$$\int_{a=0}^{b=2} \frac{dx}{x+1} = \frac{2}{2}\left(1 + \frac{1}{3}\right) = 1.333$$

$$\int \frac{dx}{x+1} = \ln|x+1|\Big]_0^2 = 1.0999$$

$$\text{error} = |1.0999 - 1.333| = 0.23477$$

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1*] Composite Trapezoidal Method



$$\int_a^b F(x)\,dx = \frac{h}{2}\left[F(a) + F(b) + 2\sum_{1}^{m-1} F(x_i)\right]$$ ... m Composite Trap Rule

$$h = \frac{b-a}{m}$$

Example: approximate $\int_0^1 e^{2x}\,dx$ using 2-Composite Trapezoidal Rule

solû: $m = 2$, $h = \frac{1-0}{2} = 0.5$, $a = 0$, $x_1 = 0.5$, $x_2 = 1$

$$\int_0^1 e^{2x}\,dx = \frac{0.5}{2}\left[F(0) + F(1) + 2F(0.5)\right]$$

$$= 3.456$$

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2] Simpson's 1/3 Rule :-

$F(x) = P_n(x)$ ...> $P_n(x)$ is 2nd order $\xrightarrow{n=2}$ Lagrange interpolation

$$\therefore \int_a^b F(x)\,dx = h/3 \left[F(x_0) + 4F(x_1) + F(x_2) \right]$$

* اشتقاق القانون نفس طريقة Trapezoidal

...> Simpson 1/3 Rule

where $h = \frac{b-a}{2}$

Example: use Simpson's rule to approximate $\int_0^6 x^2\,dx$

solu: $h = \frac{6-0}{2} = 3$, $x_0 = 0$ $x_1 = 3$ $x_2 = 6$

$$\int_0^6 x^2\,dx = \frac{3}{3}\left[F(0) + 4F(3) + F(6) \right]$$

$$= 72$$

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2*] Composite Simpson's 1/3 Rule

* عدد الفترات الفرعية داخل كل (عدد الفترات الكلي)

$$\int_a^b F(x)\,dx = \frac{h}{3}\left[F(a) + 2\sum_{i=1}^{m-1} F(x_{i-1}) + 4\sum_{i=1}^{m} F(x_{i-1}) + F(b)\right] \quad \text{...} 2m \text{ Composite Simpson's rule}$$

(2 × عدد الفترات الفرعية)

(4 × عدد الفترات الفرعية)

$$h = \frac{b-a}{2m} \quad / \quad x_i = x_0 + ih$$

No No No No No No No No No No

Example: use $2m = 6$-Composite Simpson's Rule to approximate $\int_1^{13} x^3\,dx$

solú: $2m = 6$ $\quad h = \frac{13-1}{6} = 2$

$x_0 = 1 \quad x_1 = 3 \quad x_2 = 5 \quad x_3 = 7 \quad x_4 = 9 \quad x_5 = 11$

$x_6 = 13$

$$\int x^3\,dx = \frac{2}{3}\left[F(1) + 2\left(F(5) + F(9) + 4\left(F(3) + F(7) + F(11)\right) + F(13)\right]\right] = \square$$

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3] Three - Eight Simpson's Rule

$F(x) \approx P_n(x)$ ⟶ $P_n(x)$ is 3rd order Lagrange interpolation

$n = 3$

$$\int_a^b F(x)\,dx = \frac{3h}{8}\left[F(x_0) + 3\left(F(x_1) + F(x_2)\right) + F(x_3)\right]$$ ⟶ Three - eight Simpson's rule

where $h = \frac{b-a}{3}$

Example: use (3-8) Simpson's rule to approximate $\int_1^7 x^4\,dx$

$$h = \frac{7-1}{3} = 2$$

$x_0 = 1 \quad x_1 = 3 \quad x_2 = 5 \quad x_3 = 7$

$$\int_a^b x^4\,dx = \frac{3(2)}{8}\left[F(1) + 3\left(F(3) + F(5)\right) + F(7)\right]$$

$$= 3390$$

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Initial Value Problem

فكرة هذا الجزء صعبة وانو بيعطيك إشتقاق
بدون ما انت بتقرب بقرانه لاقي اشتقاق

Euler's Method

For the equation $\frac{dy}{dt} = F(t, y)$, $a \le t \le b$, $y(a) = x$

* Consider aunifer partition , then from tylor poly. we have:

$t_0, t_1, t_2 \dots t_n$ (a, b)

$y(t) = y(t_i) + y'(t_i)(t - t_i)$

at Point t_{i+1} : $y(t_{i+1}) = y(t_i + h) = y(t_i) + y'(t_i)h$
$= y(t_i) + F(t_i, y_i)h$

y_i :

$w_0 = y_0 = y(a)$ where: $h = \frac{b-a}{n}$ ← عدد الخطوات

$w_{i+1} = w_i + h F(t_i, y_i)$ $w_i = y_i = y(t_i)$

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Example: Consider the equation $y' = -y + t + 1$, $0 \overset{a}{\leq} t \overset{b}{\leq} 1$
$y(0) = 1$ (w_0), Suppose $h = 0.2$, approximate $y(0.4)$

Solu: $t_0 = 0$, $t_1 = 0 + 0.2(1) = 0.2$, $t_2 = 0.4$

$F(t, w) = -w + t + 1$, $w_0 = 1$

$\therefore w_1 = w_0 + h F(t_0, w_0)$

$\rightsquigarrow w_1 = 1 + 0.2(-1 + 0 + 1) = 1 \approx y(0.2)$

$\therefore w_2 = w_1 + h F(t_1, w_1)$

$\rightsquigarrow w_2 = 1 + 0.2(-1 + 0.2 + 1) = 1.04 \approx y(0.4)$

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2] 2nd order Rung-Kutta Method

Assuming uniform partition to solve $\frac{dy}{dt} = F(t, y)$, $y(t_0) = \alpha$

- $y_{i+1} = y_i + (a_1 K_1 + a_2 K_2) h$
- $K_1 = F(t_i, y_i)$ / $K_2 = F(t_i + ph , y_i + q K_1 h)$
- $a_1 + a_2 = 1$ / $a_2 p = 1/2$ / $a_2 q = 1/2$

* we have three Case depend on a_2:

Case 1: Huen Method ...> $a_2 = 1/2$

Case 2: Mid Point Method ...> $a_2 = 1$

Case 3: Ralston's Method ...> $a_2 = 2/3$

* تعتمد y_{i+1} / p / q على قيمة a_2
لذلك تتغير بعلاقات في كل Case لتصبح كما سنوضح دومًا

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Case 1: Hune Method $(a_2 = \frac{1}{2})$

- $a_2 = \frac{1}{2}$ $\quad a_1 = \frac{1}{2}$ $\quad p = 1$ $\quad q = 1$
- $y_{i+1} = y_i + \frac{1}{2}(K_1 + K_2)h$
- $K_1 = F(t_i, y_i)$ $\quad K_2 = F(t_i + h, y_i + K_1 h)$

Case 2: Mid Point Method $(a_2 = 1)$

- $a_2 = 1$ $\quad a_1 = 0$ $\quad p = \frac{1}{2}$ $\quad q = \frac{1}{2}$
- $y_{i+1} = y_i + K_2 h$
- $K_1 = F(t_i, y_i)$ $\quad K_2 = F(t_i + \frac{h}{2}, y_i + \frac{K_1 h}{2})$

Case 3: Ralston's Method $(a_2 = 2/3)$

- $a_2 = 2/3$ $\quad a_1 = 1/3$ $\quad p = 3/4$ $\quad q = 3/4$
- $y_{i+1} = y_i + (\frac{K_1}{3} + \frac{2K_2}{3})h$
- $K_1 = F(t_i, y_i)$ $\quad K_2 = F(t_i + \frac{3h}{4}, y_i + \frac{3K_1 h}{4})$

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Example: Consider the initial Value Problem $y' = -y^2$, $y(t_0) = y(0) = 1$, assume $h = 0.1$, Predict $y(0.2)$ using Hune Method

* الحل لباقي الطرق يكون بنفس الطريقة

solû: $t_0 = 0$, $t_1 = 0 + h = 0.1$, $t_2 = 0.2$

$y(0.2) = y(t_2) = y_2$, $F(t, y)$

For $i = 0 \dots \rightarrow K_1 = F(t_0, y_0) = -(1)^2 = -1$

$K_2 = (0 + 0.1, 1 + -1 * 0.1) = F(0.1, 0.9) = -0.81$

$y_1 = 1 + \frac{1}{2}(K_1 + K_2)h = 0.9095$

For $i = 2 \dots \rightarrow K_1 = F(t_1, y_1) = -0.8272$

$K_2 = F(0.1, y_1 + K_1 h) = -0.6835$

$y_2 = 0.9095 + \frac{1}{2}(K_1 + K_2)h = 0.83396$

تم بحمد الله

كانت معنا صديقكم
عبد الله الكسواني

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* Finite Difference Method

Consider the Boundary Value Problem

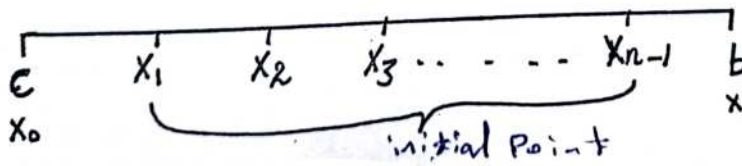
$$\frac{d^2T}{dx^2} + h_1(T_a - T) = 0 \quad \rightarrow Eq(*) \rightarrow \text{اعتبر شكل بعادية}$$

, $L = b - c$

$T(x_0) = T(c) = T_0 \rightarrow$ بداية

$T(x_n) = T_L \rightarrow$ نهاية

] $\rightarrow$ حدود



$\rightarrow L = b - a$

$\Delta X = x_i - x_{i-1}$

$\Delta X = \frac{b-c}{n}$

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The Solution of Equation (*)

$$-T_{i-1} + (2 + (h_1 * \Delta x^2))T_i - T_{i+1} = h_1 \Delta x^2 T_a$$

For $i=1$ to $i=n-1$

Ex: Use Finite-difference method for Boundary value Problem

$$T''(x) + (\underbrace{0.01}_{h_1})(\underbrace{20}_{T_a} - T(x)) = 0$$

with $\Delta x = 2$, $L = 10$, $T_0 = 40$, $T_L = 200$

Sol:

$$n = \frac{L}{\Delta x} = \frac{10}{2} = 5 \quad \Big| \quad h_1 = 0.01, \; T_a = 20$$

$T(x_0) = 40$, $T(x_5) = T_L = 200$

For $i=1$

$$-T_0 + \underbrace{\left[2 + (\underbrace{0.01}_{h} * \underbrace{2^2}_{\Delta x^2})T_1\right]}_{2.04T_1} - T_2 = \underbrace{0.8}_{0.01*2^2*20=0.8 \; \to h_1 \Delta x^2 T_x}$$

For $i=2$

$$-T_1 + \underbrace{\left[2 + (0.01 * 2^2)T_2\right]}_{2.04T_2} - T_3 = 0.8$$

For $i=3$

$$-T_2 + [2.04T_3] - T_4 = 0.8$$

For $i=4$

$$-T_3 + 2.04T_4 - T_5 = 0.8$$

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$$\begin{bmatrix} 2.04 & -1 & 0 & 0 \\ -1 & 2.04 & -1 & 0 \\ 0 & -1 & 2.04 & -1 \\ 0 & 0 & -1 & 2.04 \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix} = \begin{bmatrix} 40.8 \\ 0.8 \\ 0.8 \\ 200.8 \end{bmatrix}$$

المعروفة I راح يطلب منك بالامتحان انك تطلع هاي
وبعدها بيكلك اوجد قيم
T_4 & T_3 & T_2 & T_1

Special Matrix يتقدر كمان على
Finite difference لانو دالتها
3diagonal Matrix يعطي

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1 ملخص قوانين الجزء بالاضبط منه بمادة

1 Differentiation solu: من taylor expantion

$P_n(x) = F(x_0) + F'(x - x_1)$

- Forword ~ $f'(x_i) = \frac{F(x_i+h) - F(x_i)}{h}$
- Backword ~ $f'(x) = \frac{f(x_i) - f(x_i-h)}{h}$
- Centerd ~ $f'(x) = \frac{f(x_i+h) - f(x_i-h)}{2h}$

$h > 0$

إن ناخذ قيمتها موجبة حتى لو كانت النقاط سالبة

$$f''(x) = \frac{f(x_i+h) + F(x_i-h) - 2F(x_i)}{h^2}$$

2 Derevative By Lagrange interpotion من Lagrange

$P_n(x) = F(x_i) L_i(x) = F(x_0) \frac{x - x_1}{x_0 - x_1} \cdots$

$P_n'(x) = \sum_{i=0}^{n} F(x_i) L_i'(x)$

- ✳ ~ Three Point Formula
  - $f'(x_0) = \frac{1}{2h}[-3F(x_0) + 4F(x_1) - F(x_2)]$
  - $f'(x_0) = \frac{1}{2h}[F(x_0+h) - F(x_0-h)]$

3 Integration $\int_{x=a}^{b} F(x)\,dx$

$\int_a^b F(x)\,dx = \sum_{i=0}^{n} F(x_i) \int_{x_0}^{x_n} L_i(x)\,dx$

First Lagrange interpolation $n=1$ ~ Trapezoidal Rule $= \frac{h}{2}[F(x_1) + f(x_0)]$

~ $h = b - a$

2nd " " $n=2$ ~ Simpson's Rule $= \frac{h}{3}[F(x_0) + 4F(x_1) + F(x_2)]$

~ $h = \frac{b-a}{2}$

3rd " " $n=3$ ~ Three-eight simpson's Rule $= \frac{3h}{8}[F(x_0) + 3(F(x_1) + F(x_2)) + F(x_3)]$

~ $h = \frac{b-a}{3}$

Composite Numerical Integration

n-com Trap. $= \frac{h}{2}[F(a) + 2\sum_{i=1}^{n-1} F(x_i) + F(b)]$

~ $h = \frac{b-a}{n}$

2m-com. Sim $= \frac{h}{3}[F(a) + 2\sum_{i=1}^{m-1} F(x_i) + 4\sum_{i=1}^{m} F(x_{i-1}) + F(b)]$

~ $h = \frac{b-a}{2m}$

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to solve $\frac{dy}{dt} = f(t, y)$

Euler's Method

$w_0 = y_0 = y(t_0) = y(a)$

$w_{i+1} = w_i + h f(t_i, w_i)$

$h = \frac{b-a}{n}$

$t_i = t_0 + ih$ ($t_0 = a$)

$f(t_i, w_i)$

2nd order Rung-Kutta Method

$y_{i+1} = y_i + (a_1 K_1 + a_2 K_2) h$

$K_1 = f(t_i, y_i)$, $K_2 = f(t_i + ph, y_i + q K_1 h)$

$a_1 + a_2 = 1$ $\quad a_2 p = \frac{1}{2}$ $\quad a_2 q = \frac{1}{2}$

Case 1: Huen Method $a_2 = \frac{1}{2}$

$a_2 = \frac{1}{2}$, $a_1 = \frac{1}{2}$, $p = 1$, $q = 1$

$y_{i+1} = y_i + \frac{1}{2}(K_1 + K_2) h$

$K_1 = f(t_i, y_i)$, $K_2 = f(t_i + ph, y_i + q K_1 h)$

Case 2: Mid Point Method $a_2 = 1$

$a_2 = 1$, $a_1 = 0$, $p = \frac{1}{2}$, $q = \frac{1}{2}$

$y_{i+1} = y_i + K_2 h$

$K_1 = f(t_i, y_i)$, $K_2 = f(t_i + \frac{1}{2}h, y_i + \frac{1}{2} K_1 h)$

Case 3: Ralston's Method $a_2 = \frac{2}{3}$

$a_2 = \frac{2}{3}$, $a_1 = \frac{1}{3}$, $p = \frac{3}{4}$, $q = \frac{3}{4}$

$y_{i+1} = y_i + (\frac{1}{3} K_1 + \frac{2}{3} K_2) h$

$K_1 = f(t_i, y_i)$, $K_2 = f(t_i + \frac{3}{4}h, y_i + \frac{3}{4} K_1 h)$

Finite difference Method $-T_{i-1} + (2 + h\Delta x^2) T_i - T_{i+1} = h \Delta x^2 T_a$ $\quad \frac{dT}{dt} = h_1 (T_a - T)$

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